Counterexamples to Erdős Problem 1075

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Abstract

Erdős asked whether, for every fixed $r \geq 3$, there is a constant $c_r > r^{-r}$ such that, for every $\varepsilon > 0$, every sufficiently large r-uniform hypergraph on N vertices with at least $(1+\varepsilon)(N/r)^r$ edges contains a subgraph on $m \to \infty$ vertices with at least $c_r m^r$ edges. We give counterexamples for every $r \geq 5$. The base construction is a sequence of explicit 5-uniform hypergraphs G_n whose unnormalized Lagrangians satisfy

$$5^{-5}\left(1 + \frac{1}{4(3n)^3}\right) \leq \lambda(G_n) \leq 5^{-5} + \frac{1}{4!n}.$$

Blow-ups give the counterexample for $r = 5$, and adjoining common vertices to every edge lifts it to all larger uniformities. Thus Problem 1075, as a statement for all $r \geq 3$, has a negative answer. The endpoint question remains open for $3 \leq r \leq 4$, including the case 2/9 for 3-graphs.

1 Introduction

All hypergraphs in this paper are finite and simple. For a hypergraph H, write $v(H) = |V(H)|$, $e(H) = |E(H)|$, and let $H[S]$ denote the subgraph induced by $S \subseteq V(H)$.

Erdős proved that, for every fixed $r \geq 2$ and $\varepsilon > 0$, an r-uniform hypergraph on N vertices with at least εN^r edges contains a complete r-partite r-uniform hypergraph with equal part sizes tending to infinity with N [1]. In particular, it contains a subgraph on $m \to \infty$ vertices with $(m/r)^r$ edges. Erdős subsequently conjectured that, for each fixed $r \geq 3$, there is a constant $c_r > r^{-r}$ such that, for every $\varepsilon > 0$ and all sufficiently large N, the assumption

$$e(H) \geq (1+\varepsilon)(N/r)^r$$

forces a subgraph on $m = m(N) \to \infty$ vertices with at least $c_r m^r$ edges [3, p. 80]. This is Problem 1075 in Bloom's collection [13].

Use the density $d(H) = e(H)/\binom{v(H)}{r}$. A number $\alpha \in [0, 1)$ is a *jump* for r-graphs if there is a $c > 0$ such that, for every $\varepsilon > 0$ and integer $m \geq r$, all sufficiently large r-graphs of density at least $\alpha + \varepsilon$ contain an m-vertex subgraph of density at least $\alpha + c$; otherwise it is a *non-jump* [4]. The threshold $(N/r)^r$ corresponds asymptotically to $\alpha_r = r!/r^r$. Erdős proved that every number in $[0, \alpha_r)$ is a jump for r-graphs [2]. Thus Problem 1075 is the endpoint question at α_r. Frankl and Rödl disproved the earlier jumping constant conjecture by constructing non-jumps for every $r \geq 3$ [4, Theorem 1.2]. Frankl, Peng, Rödl and Talbot later showed that $(5/2)r!/r^r$ is a non-jump for every $r \geq 3$ [5], and Peng proved the lifting theorem that carries a non-jump of the form $c\,r!/r^r$ to all larger uniformities [6, Theorem 1]. Yan and Peng proved that 12/25 is a non-jump for 3-graphs, yielding $(54/25)r!/r^r$ for every $r \geq 3$ by lifting [7].

Shaw recently proved that $2r!/r^r$ is a non-jump for every $r \geq 4$, and that for $r \geq 4$ no smaller non-jump can be obtained from his finite-pattern formulation of the Frankl–Rödl method [8, Theorems 4.2 and 4.3]. Liu and Mubayi subsequently proved that 4/9 is a non-jump for 3-graphs [9]; this lies below the threshold $6(5\sqrt{5}-2)/121$ in Shaw's Theorem 4.3(i). The endpoint 2/9 for 3-graphs remains open; Baber and Talbot proved, using flag algebras, the first jump intervals immediately above it [10]. The result below shows that the endpoint is a non-jump for every $r \geq 5$; the cases $3 \leq r \leq 4$ remain open.

The construction below is not an instance of Shaw's finite-pattern criterion. Its distinguished links between the two vertex families follow the matching A_iB_i and the shifted matching B_iA_{i+1} around a cycle, and the proof controls the full Lagrangian of a growing sequence G_n.

Theorem 1. *For every $\gamma > 5^{-5}$, there is an $\varepsilon > 0$ and there are arbitrarily large 5-uniform hypergraphs H such that*

$$e(H) \geq (1+\varepsilon)\big(v(H)/5\big)^5, \qquad e(H[S]) < \gamma |S|^5 \quad \text{for every nonempty } S \subseteq V(H).$$

The conclusion is stronger than the formulation of Problem 1075, since the bound holds for every nonempty vertex subset, and any subgraph on that vertex set has at most as many edges as the induced subgraph. Taking $5^{-5} < \gamma < c_5$ in Theorem 1 rules out any $c_5 > 5^{-5}$. In the usual density normalization, it also implies that $5!/5^5$ is a non-jump. The proof uses explicit finite hypergraphs whose Lagrangians approach 5^{-5} from above. We use the unnormalized Lagrangian of [4, Section 2],

$$P_H(x) = \sum_{e \in E(H)} \prod_{v \in e} x_v, \qquad \lambda(H) = \max_{\substack{x_v \geq 0 \\ \sum_v x_v = 1}} P_H(x), \tag{1.1}$$

without the factor $r!$; see also [11, 12].

Proposition 2. *For every integer $n \geq 2$, the 5-uniform hypergraph G_n defined in Section 2 has $2n+7$ vertices and satisfies*

$$5^{-5}\left(1 + \frac{1}{4(3n)^3}\right) \leq \lambda(G_n) \leq 5^{-5} + \frac{1}{4!n}. \tag{1.2}$$

The construction seeks a density gain that requires the entire cycle. The two alternating matchings allow the balancing conditions to hold simultaneously around a closed cycle, leaving room for an additional edge to increase the Lagrangian. Opening the cycle creates a terminal mismatch that absorbs this gain. Since a long cycle always has a vertex of small weight, the passage from a cycle to a path also gives an upper bound approaching the same threshold.

The mechanism is visible at the level of the cubic forms. After normalizing the bottom mass, the balanced weighting on the closed cycle has

$$b_i + c = B - b_i + D = a_{i+1} + c = A - a_{i+1} + D = \frac{1}{3},$$

so both cubic forms attain 1/27 even with $D > 0$, while the additional edge $\{U, V\} \cup \mathcal{D}$ contributes a gain of order D^3. Deleting one vertex A_k opens the cycle; the terminal mismatch forces a loss of order D^3 with a coefficient large enough to absorb that gain. Consequently every open path T_L has Lagrangian exactly 5^{-5}, whereas the closed cycle has Lagrangian slightly above 5^{-5}.

2 The cyclic construction

Fix $n \geq 2$. Take pairwise distinct vertices

$$A_i, B_i \quad (i \in \mathbb{Z}/n\mathbb{Z}), \qquad C, U, V, \qquad W, \qquad D_1, D_2, D_3.$$

Put $\mathcal{D} = \{D_1, D_2, D_3\}$. On the vertex set $\{A_i, B_i : i \in \mathbb{Z}/n\mathbb{Z}\} \cup \{C\} \cup \mathcal{D}$, define two three-uniform hypergraphs by

$$E(H_1) = \bigcup_i \{\{A_i, X, Y\} : X \in \{B_i, C\},\ Y \in \{B_j : j \neq i\} \cup \mathcal{D}\},$$
$$E(H_2) = \bigcup_i \{\{B_i, X, Y\} : X \in \{A_{i+1}, C\},\ Y \in \{A_j : j \neq i+1\} \cup \mathcal{D}\}.$$

All subscripts are read modulo n. Define

$$\begin{aligned} E(G_n) = &\{\{W, U\} \cup e : e \in E(H_1)\} \\ &\cup \{\{W, V\} \cup e : e \in E(H_2)\} \cup \{\{U, V\} \cup \mathcal{D}\}. \end{aligned} \tag{2.1}$$

Every edge has five distinct vertices. The three families are distinguished by their intersections with $\{U, V\}$, and within each of the first two families the triple e determines the edge uniquely. Thus G_n is a simple 5-uniform hypergraph.

Use lower-case letters for the weights of the corresponding vertices, and write

$$A = \sum_i a_i, \qquad B = \sum_i b_i, \qquad D = \sum_{j=1}^{3} d_j.$$

The two cubic forms and the polynomial of G_n are

$$P_1 = \sum_i a_i(b_i + c)(B - b_i + D), \tag{2.2}$$

$$P_2 = \sum_i b_i(a_{i+1} + c)(A - a_{i+1} + D), \tag{2.3}$$

$$P_{G_n} = w(uP_1 + vP_2) + uv \prod_{j=1}^{3} d_j. \tag{2.4}$$

Lemma 3. *The lower bound in* (1.2) *holds.*

Proof. Assign the weights

$$\begin{gathered} w = \frac{1}{5}, \qquad u = v = \frac{1}{10}, \qquad a_i = b_i = \frac{1}{5n}, \\ d_j = \frac{1}{15n}, \qquad c = \frac{n-1}{5n}. \end{gathered} \tag{2.5}$$

Their sum is one. Moreover,

$$A = B = c + D = \frac{1}{5}, \qquad b_i + c = B - b_i + D = a_{i+1} + c = A - a_{i+1} + D = \frac{1}{5}.$$

Thus $P_1 = P_2 = 5^{-3}$. The first term in (2.4) is 5^{-5}, and the second is

$$\frac{1}{100(15n)^3} = \frac{5^{-5}}{4(3n)^3}. \qquad \square$$

3 A bound for open paths

The path estimate uses the following inequality for finite sequences.

Lemma 4. *Let $N \geq 1$, and let $x_0, \ldots, x_N$ be nonnegative real numbers such that $x_N = 0$ and $\sum_{i=0}^{N-1} x_i \geq 1/2$. Then*

$$\sum_{i=0}^{N-1} x_i(1 - x_{i+1})^2 \geq \frac{2}{9}. \tag{3.1}$$

Proof. Define

$$\psi(x) = \begin{cases} x - x^2/2, & 0 \leq x \leq 1, \\ 1/2, & x > 1. \end{cases}$$

For all $x, y \geq 0$,

$$\psi(x) - \psi(y) \leq x(1 - y)^2. \tag{3.2}$$

For $x, y \leq 1$, this follows from the identity

$$x(1-y)^2 - \psi(x) + \psi(y) = \frac{(x - 2y + y^2)^2 + y(1-y)^2(2-y)}{2} \geq 0.$$

If $y \geq 1$, the left side of (3.2) is nonpositive. If $x > 1 > y$, it equals $(1-y)^2/2$, so the inequality holds in this case as well.

If some $x_j \geq 1/3$, summing (3.2) from j to $N-1$ gives

$$\sum_{i=0}^{N-1} x_i(1 - x_{i+1})^2 \geq \psi(x_j) - \psi(0) \geq \frac{5}{18} > \frac{2}{9}.$$

Otherwise every $x_i < 1/3$, and the sum is at least $\frac{4}{9} \sum_{i=0}^{N-1} x_i \geq 2/9$. $\square$

Lemma 5. *Let $L \geq 1$, and let $a_1, \ldots, a_L, b_1, \ldots, b_L, c, D$ be nonnegative numbers. Put*

$$A = \sum_{i=1}^{L} a_i, \qquad B = \sum_{i=1}^{L} b_i, \qquad a_{L+1} = 0,$$

and suppose that $A + B + c + D = 1$. Define P_1, P_2 by (2.2) and (2.3), with sums over $1 \leq i \leq L$. Then, for every $s \in [0, 1]$,

$$sP_1 + (1-s)P_2 \leq \frac{1}{27} - \frac{s(1-s)D^3}{10}. \tag{3.3}$$

Proof. Set

$$\delta_A = A - \frac{1}{3}, \qquad \delta_B = B - \frac{1}{3}, \qquad S = |\delta_A| + |\delta_B|,$$

and

$$\alpha = \frac{A + D - c}{2} = D + \delta_A + \frac{\delta_B}{2}, \qquad \beta = \frac{B + D - c}{2} = D + \frac{\delta_A}{2} + \delta_B.$$

Completing the square gives

$$P_1 = \frac{A(1-A)^2}{4} - \sum_i a_i(b_i - \beta)^2, \tag{3.4}$$

$$P_2 = \frac{B(1-B)^2}{4} - \sum_i b_i(a_{i+1} - \alpha)^2. \tag{3.5}$$

Write

$$E = s\sum_i a_i(b_i-\beta)^2 + (1-s)\sum_i b_i(a_{i+1}-\alpha)^2, \qquad \Delta = \frac{1}{27} - sP_1 - (1-s)P_2.$$

The exact identity

$$\Delta = \frac{s\delta_A^2(4/3-A)}{4} + \frac{(1-s)\delta_B^2(4/3-B)}{4} + E$$

and the bounds $A, B \le 1-D$, $D \le 1$ imply

$$\Delta \ge \frac{D}{3}\left(s\delta_A^2 + (1-s)\delta_B^2\right) + E.$$

Indeed, $(4/3-A)/4 \ge (1/3+D)/4 \ge D/3$, and similarly for B. By the weighted Cauchy–Schwarz inequality,

$$\Delta \ge \frac{s(1-s)DS^2}{3} + E. \tag{3.6}$$

If $S \ge 11D/20$, then

$$\Delta \ge \frac{121}{1200}s(1-s)D^3 \ge \frac{s(1-s)D^3}{10},$$

as required. Suppose therefore that $S < 11D/20$. Since $D \le 1/3 + S$, we have

$$D \le \frac{20}{27}, \qquad A+B \ge \frac{2}{3} - S \ge \frac{7}{27} > \frac{1}{4}, \qquad 0 < D-S \le \alpha, \beta \le \frac{1}{2}.$$

Apply Lemma 4 to the sequence

$$\frac{a_1}{\alpha}, \frac{b_1}{\beta}, \ldots, \frac{a_L}{\alpha}, \frac{b_L}{\beta}, 0.$$

Its sum is $A/\alpha + B/\beta \ge 2(A+B) > 1/2$. Each successive pair x, y contributes to E a term $s\alpha\beta^2 x(1-y)^2$ or $(1-s)\beta\alpha^2 x(1-y)^2$. Consequently

$$E \ge \frac{2}{9}\min\{s\alpha\beta^2, (1-s)\beta\alpha^2\} \ge \frac{2}{9}s(1-s)(D-S)^3. \tag{3.7}$$

For all $S, m \ge 0$ we have

$$\frac{(S+m)S^2}{3} + \frac{2m^3}{9} \ge \frac{(S+m)^3}{10}. \tag{3.8}$$

Multiplying the difference by 90 gives the nonnegative expression

$$\begin{aligned} &21S^3 + 3S^2m - 27Sm^2 + 11m^3 \\ &\quad = 21\left(S - \frac{3m}{5}\right)^2\left(S + \frac{6m}{5}\right) + 3m\left(S - \frac{18m}{25}\right)^2 + \frac{233m^3}{625}. \end{aligned}$$

Taking $m = D - S$ in (3.8) and using (3.6)–(3.7) proves (3.3). $\square$

For $L \ge 1$, define the 5-uniform hypergraph T_L as in (2.1), with each index ranging over $1, \ldots, L$ and with the last link in H_2 replaced by the triples

$$\{B_L, C, Y\}, \qquad Y \in \{A_1, \ldots, A_L\} \cup \mathcal{D}.$$

The edge $\{U, V\} \cup \mathcal{D}$ is retained. Its polynomial is (2.4) with $a_{L+1} = 0$.

Lemma 6. *For every $L \geq 1$, we have $\lambda(T_L) = 5^{-5}$.*

Proof. The lower bound follows by assigning weight $1/5$ to the vertices of any one edge and zero elsewhere. For the upper bound, take nonnegative vertex weights of total mass one, and put

$$h = w, \qquad q = u + v, \qquad t = A + B + c + D.$$

Thus $h + q + t = 1$. If $q = 0$ or $t = 0$, then $P_{T_L} = 0$. Otherwise put

$$s = u/q, \qquad d = D/t, \qquad z = s(1-s)d^3 \in [0, 1/4].$$

Applying Lemma 5 to the bottom weights divided by t, and then using homogeneity, gives

$$uP_1 + vP_2 \leq qt^3 \left(\frac{1}{27} - \frac{z}{10}\right).$$

The arithmetic–geometric mean inequality now yields

$$P_{T_L} \leq hqt^3 \left(\frac{1}{27} - \frac{z}{10}\right) + q^2 s(1-s) \left(\frac{td}{3}\right)^3. \tag{3.9}$$

Also,

$$\frac{qt^3}{27} \leq \left(\frac{q+t}{4}\right)^4, \qquad \frac{q^2t^3}{3^3} \leq 4\left(\frac{q+t}{5}\right)^5. \tag{3.10}$$

Define

$$F(h) = \frac{3125}{256} h(1-h)^4.$$

We have $F(h) \leq 1$, with equality at $h = 1/5$. Since $1 - 27z/10 > 0$, we may substitute (3.10) into (3.9). We get

$$\begin{aligned} \frac{P_{T_L}}{5^{-5}} &\leq F(h)\left(1 - \frac{27}{10}z\right) + 4s(1-s)d^3(1-h)^5 \\ &= F(h) + z\left[4(1-h)^5 - \frac{27}{10}F(h)\right]. \end{aligned} \tag{3.11}$$

It remains to show that

$$\left(1 - \frac{27}{40}\right) F(h) + (1-h)^5 \leq 1 \qquad (0 \leq h \leq 1). \tag{3.12}$$

Indeed, the last expression in (3.11) is the convex combination

$$(1-4z)F(h) + 4z\left[\left(1 - \frac{27}{40}\right)F(h) + (1-h)^5\right].$$

Since $(1 - 27/40)(3125/256) = 8125/2048 < 5$, the binomial expansion gives

$$\left(1 - \frac{27}{40}\right) F(h) + (1-h)^5 \leq 5h(1-h)^4 + (1-h)^5 \leq \big(h + (1-h)\big)^5 = 1.$$

This completes the upper bound. $\square$

4 The Lagrangian bound and the blow-ups

Proof of Proposition 2. The lower bound was proved in Lemma 3. For the upper bound, take any nonnegative weights on G_n of total mass one. Choose an index k with

$$a_k \le \frac{A}{n} \le \frac{1}{n}.$$

After deleting A_k, put $a'_1 = a'_{n+1} = 0$, $a'_j = a_{k+j-1}$ for $2 \le j \le n$, and $b'_j = b_{k+j-1}$ for $1 \le j \le n$, with indices modulo n. Writing $A' = \sum_j a'_j$, $B' = \sum_j b'_j$, the remaining polynomial is

$$\begin{aligned} P_{G_n - \{A_k\}} &= wu \sum_{j=1}^{n} a'_j (b'_j + c)(B' - b'_j + D) \\ &\qquad + wv \sum_{j=1}^{n} b'_j (a'_{j+1} + c)(A' - a'_{j+1} + D) + uv \prod_{\ell=1}^{3} d_\ell \\ &= P_{T_n}\big|_{a'_1 = 0}. \end{aligned} \tag{4.1}$$

Thus Lemma 6 and homogeneity bound the total weight of the edges that remain by

$$5^{-5}(1 - a_k)^5 \le 5^{-5}.$$

For any simple 5-uniform hypergraph and weights of total mass one,

$$\frac{\partial P_H}{\partial x_v} \le \sum_{\substack{S \subseteq V(H) \setminus \{v\} \\ |S| = 4}} \prod_{w \in S} x_w \le \frac{(\sum_{w \ne v} x_w)^4}{4!} \le \frac{1}{4!}. \tag{4.2}$$

Indeed, each product of four distinct weights occurs 4! times in the power expansion, and all other terms are nonnegative. The deleted edges consequently have total weight at most

$$a_k \frac{\partial P_{G_n}}{\partial a_k} \le \frac{a_k}{4!} \le \frac{1}{4!n}.$$

This proves (1.2). □

Proof of Theorem 1. Fix $\gamma > 5^{-5}$, and choose an integer $n \ge 2$ such that

$$5^{-5} + \frac{1}{4!n} < \gamma.$$

Keep n fixed, and set

$$\varepsilon = \frac{1}{4(3n)^3} > 0.$$

For each positive integer M, form a blow-up $B_{n,M}$ of G_n: replace each vertex by an independent class, and each edge by all transversal edges across its five classes. Use the following class sizes, with one class for each listed vertex:

vertex	W	U, V	A_i, B_i	C	D_j
class size	$6nM$	$3nM$	$6M$	$6(n-1)M$	$2M$

There are $N = 30nM$ vertices. These class sizes are proportional to the weights in (2.5), and the computation in Lemma 3 gives the exact edge count

$$e(B_{n,M}) = \left[(6n)^5 + 72n^2\right]M^5 = (1+\varepsilon)(N/5)^5. \tag{4.3}$$

Let S be any nonempty set of m vertices of $B_{n,M}$, and let k_v be the number of its vertices in the class corresponding to $v \in V(G_n)$. Then

$$e(B_{n,M}[S]) = \sum_{e \in E(G_n)} \prod_{v \in e} k_v = m^5 P_{G_n}\big((k_v/m)_v\big) \le m^5 \lambda(G_n) < \gamma m^5.$$

Together with (4.3) and $M \to \infty$, this proves the theorem. □

Corollary 7. *For every integer $r \ge 5$ and every $\gamma > r^{-r}$, there is an $\varepsilon > 0$ and there are arbitrarily large r-uniform hypergraphs H such that*

$$e(H) \ge (1+\varepsilon)\big(v(H)/r\big)^r, \qquad e(H[S]) < \gamma |S|^r \quad \textit{for every nonempty } S \subseteq V(H).$$

In particular, Problem 1075 fails for every $r \ge 5$.

Proof. We implement Peng's lifting construction [6, proof of Theorem 1, p. 763] by adjoining vertices before taking a blow-up. Let $q = r - 5$. For a 5-uniform hypergraph G, form an r-uniform hypergraph G^{+q} by adjoining the same set of q new vertices to every edge. If the total weight on $V(G)$ is t, then the arithmetic–geometric mean inequality gives

$$P_{G^{+q}} \le \lambda(G) t^5 \left(\frac{1-t}{q}\right)^q,$$

with the usual interpretation when $q = 0$. The right side is maximized at $t = 5/r$, and equality is attained by taking an optimal weighting of G, scaled by $5/r$, and weight $1/r$ on each new vertex. Hence

$$\lambda(G^{+q}) = \frac{5^5}{r^r}\lambda(G). \tag{4.4}$$

Applying (4.4) to G_n and using Proposition 2 gives

$$r^{-r}\left(1 + \frac{1}{4(3n)^3}\right) \le \lambda(G_n^{+q}) \le r^{-r} + \frac{5^5}{r^r 4! n}.$$

Choose n so that the upper bound is less than γ. The blow-up argument in the proof of Theorem 1, using the rational weighting obtained by suspending (2.5), then gives the stated r-uniform hypergraphs with $\varepsilon = 1/[4(3n)^3]$. □

The direct construction with $r-4$ common W-vertices and $r-2$ vertices in $\mathcal{D}$ has gain of order D^{r-2}; at $r = 5$ this matches the cubic open-path loss, and Lemma 6 shows that the constants suffice to absorb it. At $r = 4$, the quadratic gain dominates this cubic loss for small positive D, so the same open-path bound no longer controls the gain. At $r = 3$, the formal gain edge would contain only one vertex of $\mathcal{D}$, giving a linear term, and the four-vertex core edges cannot be retained in a 3-uniform hypergraph. The endpoint question for $r = 3, 4$ remains open.

Together with Erdős's theorem that every number below $r!/r^r$ is a jump [2], Corollary 7 identifies $r!/r^r$ as the smallest non-jump for every $r \geq 5$. This leaves a natural question: are all numbers in $[r!/r^r, 1)$ non-jumps, or do jumps occur again above the first non-jump? The possibility of a single transition from jumps to non-jumps remains consistent with the result.

For $r = 3$, Liu and Mubayi conjecture that every number in $[0, 4/9)$ is a jump and every number in $[4/9, 1)$ is a non-jump [9, Conjecture 1.2]. If this conjecture holds, the smallest non-jump for 3-graphs would be twice the Erdős threshold, whereas for every $r \geq 5$ it equals that threshold. Understanding whether this difference between uniformities occurs is a natural next question.

Declaration of generative AI and AI-assisted technologies in the writing process

GPT-6 Astra was used to generate the mathematical proofs, draft the manuscript, and perform the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used only for editorial review of the exposition. GPT-6 Astra was run in a research environment containing earlier results produced by GPT-5.6 Sol and Claude Opus 5, but those earlier results did not contribute to the final mathematical arguments. The author reviewed the final manuscript and takes full responsibility for its content.

References

[1] P. Erdős, On extremal problems of graphs and generalized graphs, *Israel J. Math.* **2** (1964), 183–190.

[2] P. Erdős, On some extremal problems on r-graphs, *Discrete Math.* **1** (1971), 1–6.

[3] P. Erdős, Extremal problems on graphs and hypergraphs, in *Hypergraph Seminar* (Columbus, Ohio, 1972), Lecture Notes in Mathematics, vol. 411, Springer, Berlin, 1974, pp. 75–84.

[4] P. Frankl and V. Rödl, Hypergraphs do not jump, *Combinatorica* **4** (1984), 149–159.

[5] P. Frankl, Y. Peng, V. Rödl and J. Talbot, A note on the jumping constant conjecture of Erdős, *J. Combin. Theory Ser. B* **97** (2007), 204–216.

[6] Y. Peng, On jumping densities of hypergraphs, *Graphs Combin.* **25** (2009), 759–766.

[7] Z. Yan and Y. Peng, Non-jumping Turán densities of hypergraphs, *Discrete Math.* **346** (2023), 113195.

[8] B. R. Shaw, Minimal hypergraph non-jumps, *European J. Combin.* **137** (2026), 104400; first circulated as arXiv:2506.09620 (2025).

[9] X. Liu and D. Mubayi, The number 4/9 is a non-jump for 3-graphs, arXiv:2605.13567 (2026).

[10] R. Baber and J. Talbot, Hypergraphs do jump, *Combin. Probab. Comput.* **20** (2011), 161–171.

[11] T. S. Motzkin and E. G. Straus, Maxima for graphs and a new proof of a theorem of Turán, *Canad. J. Math.* **17** (1965), 533–540.

[12] P. Keevash, Hypergraph Turán problems, in *Surveys in Combinatorics 2011*, London Math. Soc. Lecture Note Ser., vol. 392, Cambridge Univ. Press, Cambridge, 2011, pp. 83–140.

[13] T. F. Bloom, *Erdős Problem #1075*, https://www.erdosproblems.com/1075. Accessed 6 September 2026.